



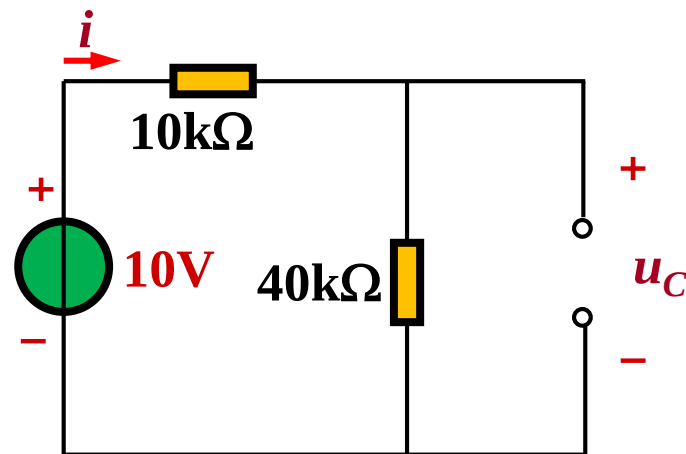
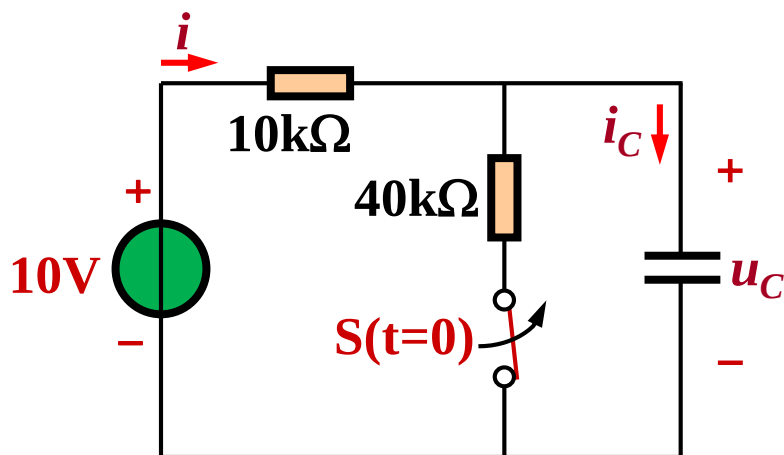
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电路的暂态响应例题

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例1 求图示电路中开关S断开后电容电流的初始值 $i_C(0_+)$ 。

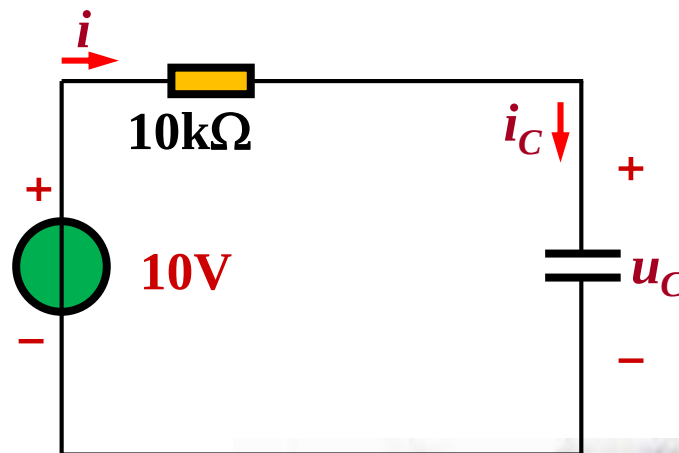


$t=0_-$ 等效电路

【解】 (1) 画 $t=0_-$ 时刻等效电路。

$$\begin{aligned} u_C(0_+) &= u_C(0_-) \\ &= \frac{40}{40+10} \times 10 = 8V \end{aligned}$$

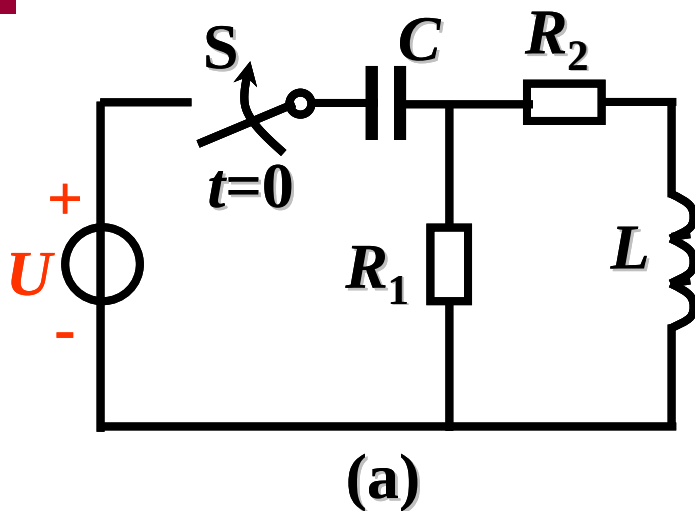
$$(2) i_C(0_+) = \frac{10-8}{10} = 0.2\text{mA}$$



$t=0_+$ 等效电路

例2

暂态过程初始值的确定



已知：换路前电路处稳态， C 、 L 均未储能。

试求：电路中各电压和电流的初始值。

解：(1)由换路前电路求 $u_C(0_-)$, $i_L(0_-)$

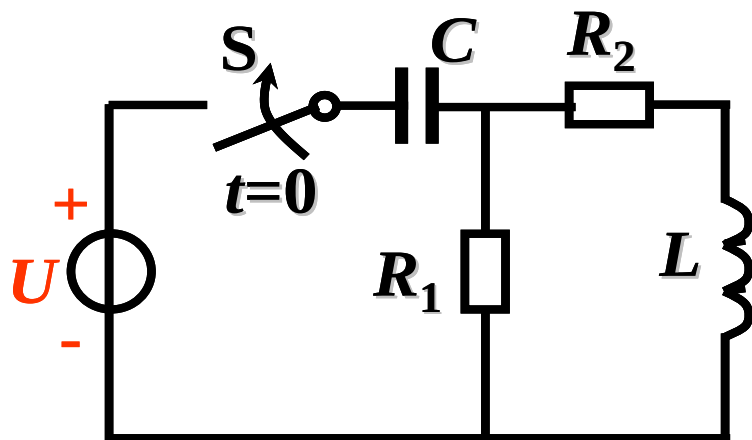
由已知条件知 $u_C(0_-) = 0$, $i_L(0_-) = 0$

根据换路定则得： $u_C(0_+) = u_C(0_-) = 0$

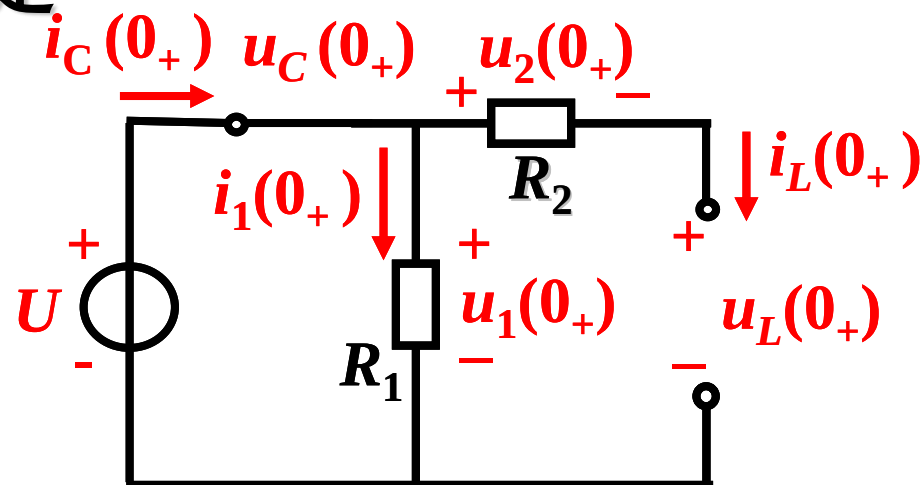
$i_L(0_+) = i_L(0_-) = 0$



暂态过程初始值的确定



(a)



(b) $t = 0_+$ 等效电路

(2) 由 $t=0_+$ 电路，求其余各电流、电压的初始值

$u_C(0_-) = 0$ ，换路瞬间，电容元件可视为短路。

$i_L(0_-) = 0$ ，换路瞬间，电感元件可视为开路。

$$i_C(0_+) = i_1(0_+) = \frac{U}{R}$$

$$u_L(0_+) = u_1(0_+) = U$$

$$(i_C(0_-) = 0)$$

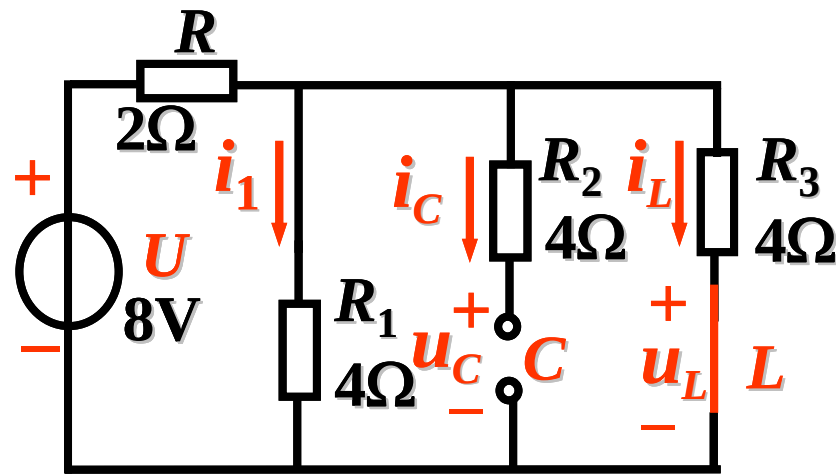
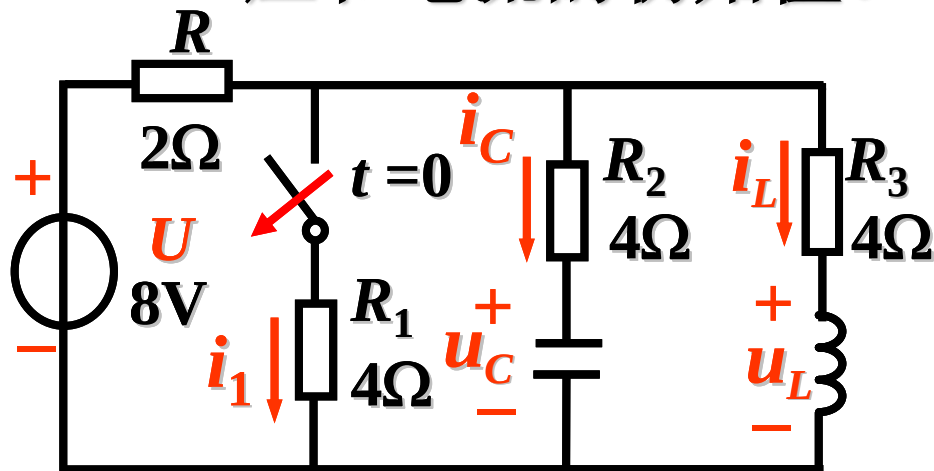
$$(u_L(0_-) = 0)$$

i_C 、 u_L 产生突变

$$u_2(0_+) = 0$$

例3

换路前电路处于稳态。试求图示电路中各个电压和电流的初始值。



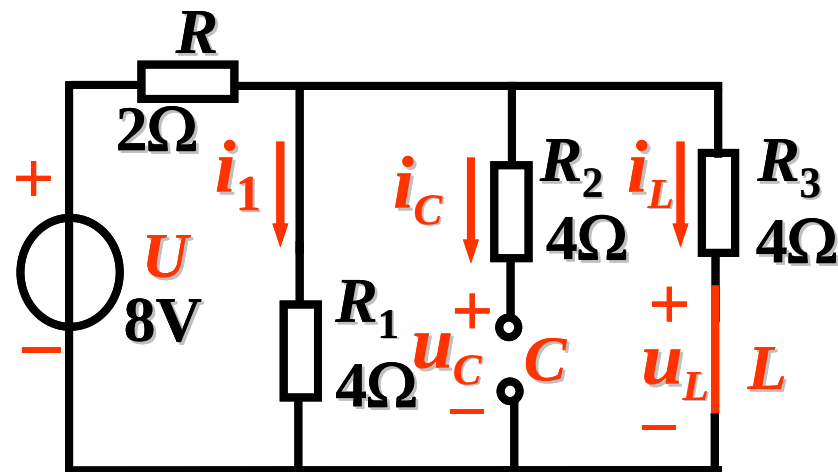
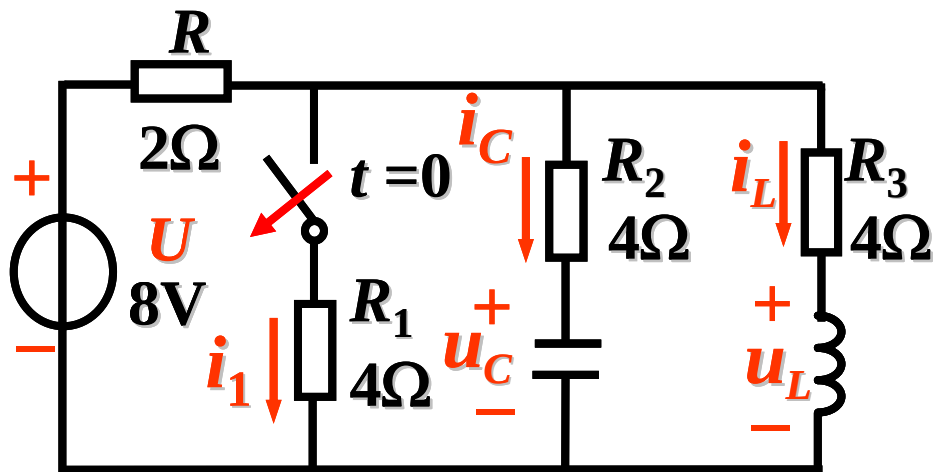
$t = 0_-$ 等效电路

解: (1) 由 $t = 0_-$ 电路求 $u_C(0_-)$ 、 $i_L(0_-)$

换路前电路已处于稳态: 电容元件视为开路;

由 $t = 0_-$ 电路可求得: 电感元件视为短路。

$$i_L(0_-) = \frac{R_1}{R_1 + R_3} \times \frac{U}{R + \frac{R_1 R_3}{R_1 + R_3}} = \frac{4}{4 + 4} \times \frac{U}{2 + \frac{4 \times 4}{4 + 4}} = 1 \text{ A}$$



$t = 0_-$ 等效电路

解: (1) $i_L(0_-) = 1\text{ A}$

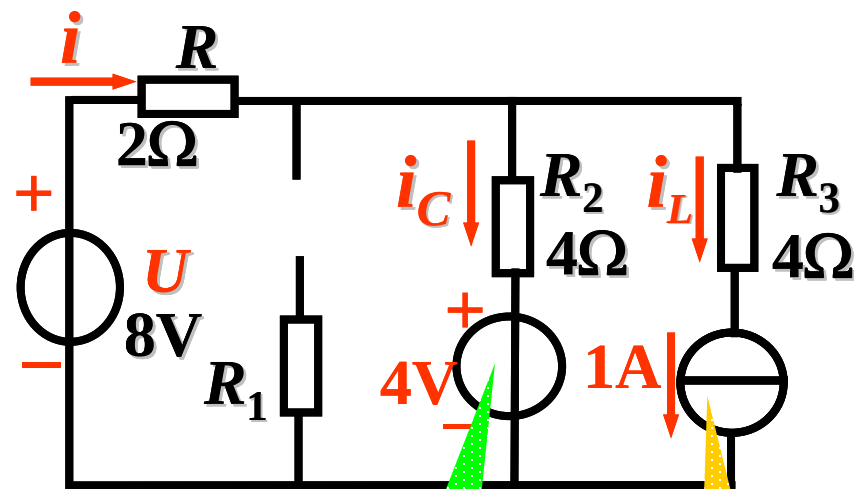
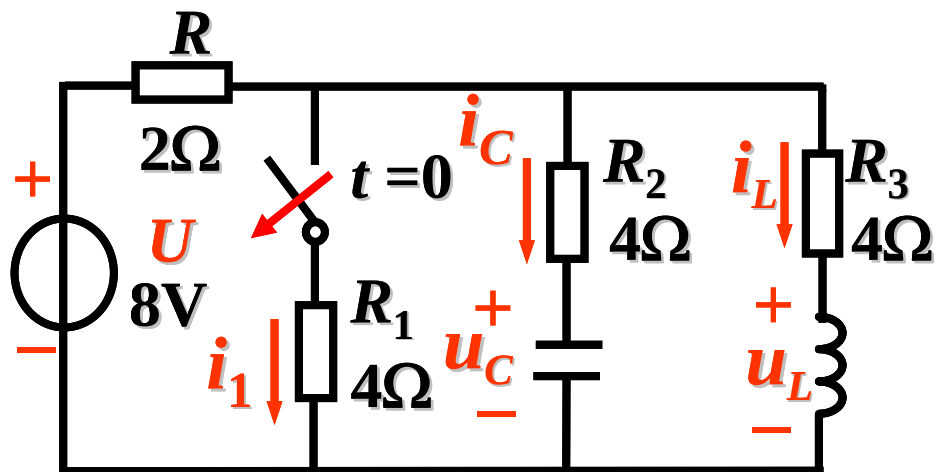
$$u_C(0_-) = R_3 i_L(0_-) = 4 \times 1 = 4\text{ V}$$

由换路定则:

$$i_L(0_+) = i_L(0_-) = 1\text{ A}$$

$$u_C(0_+) = u_C(0_-) = 4\text{ V}$$





$t = 0_+$ 时等效电路

解: (2) 由 $t = 0_+$ 电路求 $i_C(0_+)$ 、 $u_L(0_+)$ $u_C(0_+)$ $i_L(0_+)$

由图可列出 $U = Ri(0_+) + R_2 i_C(0_+) + u_C(0_+)$

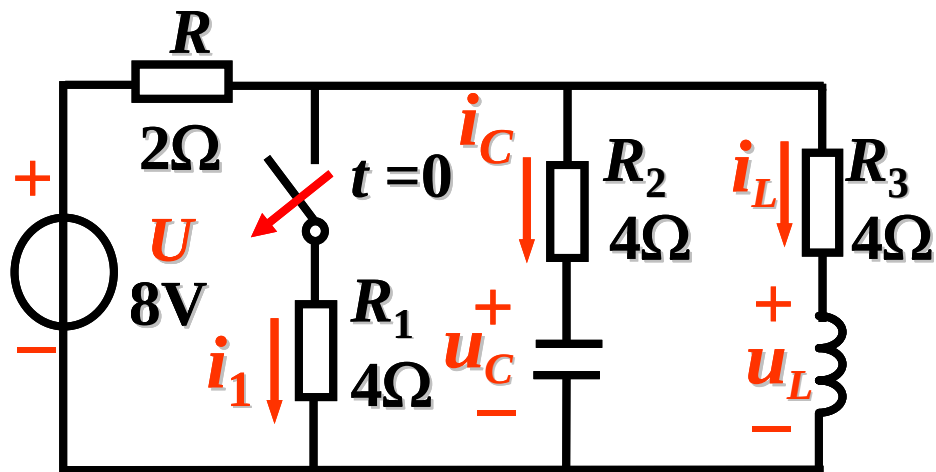
$$i(0_+) = i_C(0_+) + i_L(0_+)$$

带入数据

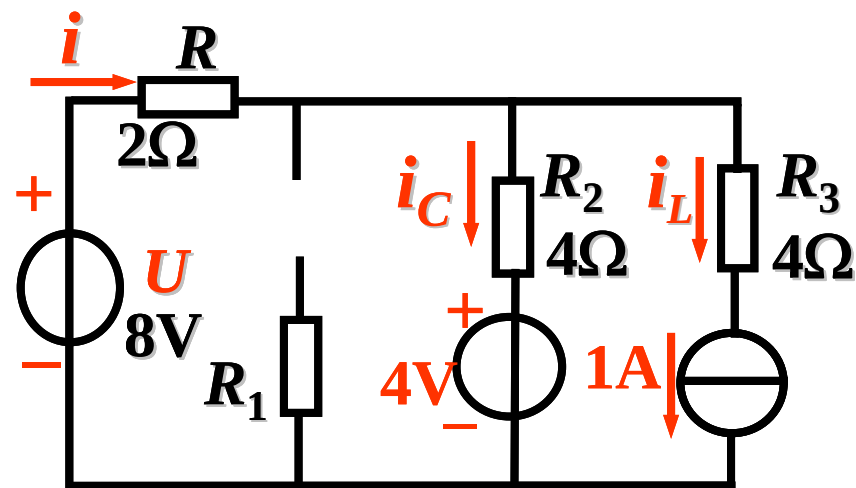
$$8 = 2i(0_+) + 4i_C(0_+) + 4$$

$$i(0_+) = i_C(0_+) + 1$$





解：解之得 $i_C(0_+) = \frac{1}{3} \text{ A}$
并可求出

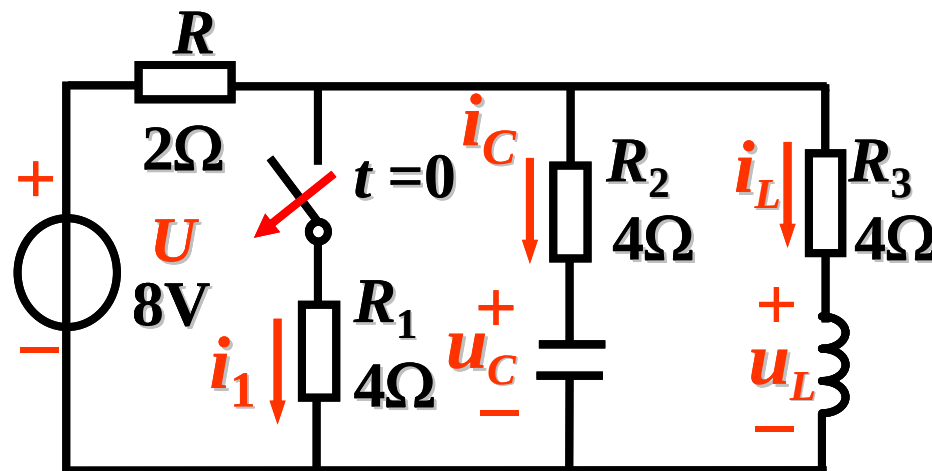


$t = 0_+$ 时等效电路

$$\begin{aligned} u_L(0_+) &= R_2 i_C(0_+) + u_C(0_+) - R_3 i_L(0_+) \\ &= 4 \times \frac{1}{3} + 4 - 4 \times 1 = 1\frac{1}{3} \text{ V} \end{aligned}$$



计算结果:



电量	u_C / V	i_L / A	i_C / A	u_L / V
$t = 0_-$	4	1	0	0
$t = 0_+$	4	1	$\frac{1}{3}$	$1\frac{1}{3}$

换路瞬间, u_C 、 i_L 不能跃变, 但 i_C 、 u_L 可以跃变。



例4 电路原处于稳态， $t=0$ 时开关 S 闭合。已知 $u_C(0_-) = 1V$ 、 $C = 10\mu F$ ，求开关闭合后的电容电压 u_C 和电流 i_C 及电流源两端的电压 u 。

【解】 (1) 三要素法求 u_C 。

$$u_C(0_+) = u_C(0_-) = 1V$$

$$u_C(\infty) = 1 \times 1 + 10 = 11V$$

$$\tau = RC = 2 \times 10^3 \times 10 \times 10^{-6} = 2 \times 10^{-2} S$$

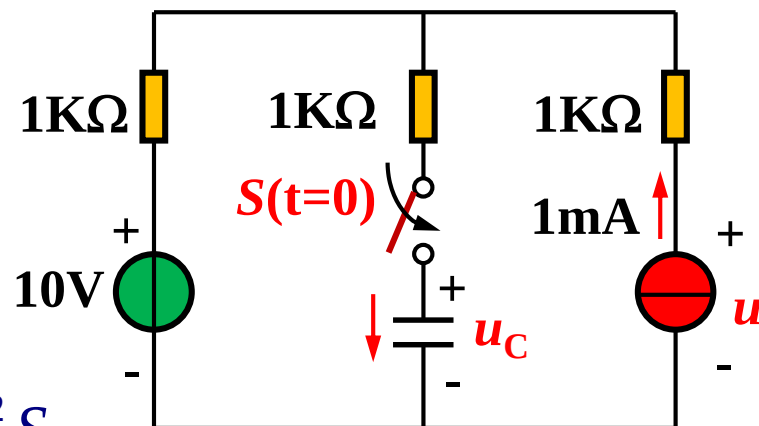
$$u_C(t) = u_C(\infty) + [u_C(0_+) - u_C(\infty)]e^{-\frac{t}{\tau}}$$

$$= 11 - 10e^{-50t}(V)(t \geq 0)V$$

(2) 其他电量 $i_C(t) = C \frac{du_C}{dt} = 5e^{-50t}(mA)(t \geq 0)$

$$u(t) = 1 \times 1 + 1 \times 5e^{-50t} + 11 - 10e^{-50t}$$

$$= 12 - 5e^{-50t}(V)(t \geq 0)$$



例5 电路原处稳态，当 $t=0$ 开关闭合，试求 $t \geq 0$ 时 $u_C(t)$ 和 $i_C(t)$ 。

【解】 (1) 用换路前电路计算

$$u_C(0_+) = u_C(0_-) = \frac{1+2}{1+1+2} \times 6 = 4.5\text{V}$$

(2) 用换路后电路计算

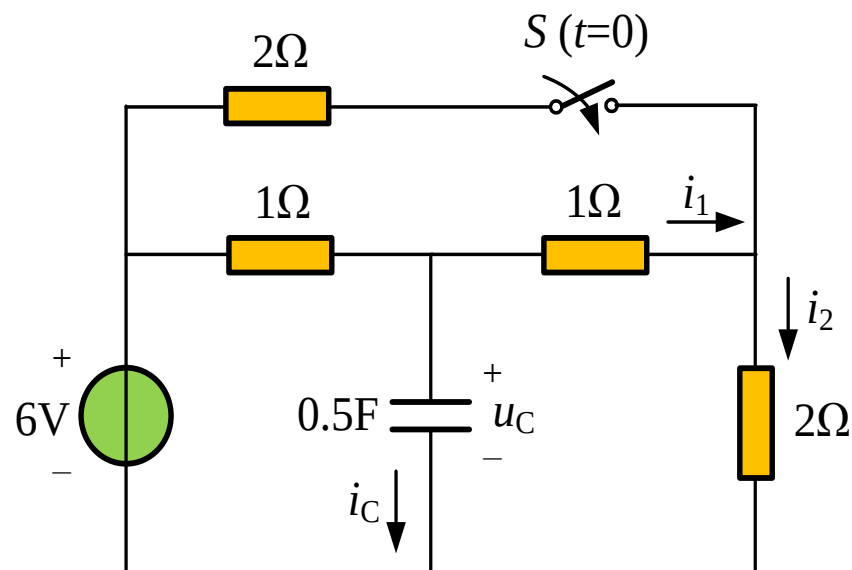
$$u_C(\infty) = \frac{6}{1+2} \times 2 + \frac{1}{1+1} \times 2 = 5\text{V}$$

(3) 用换路后电路计算

$$\tau = CR = 0.5 \times \left(\frac{2 \times 2}{2+2} + 1 \right) \parallel 1 = \frac{1}{2} \times \frac{2}{3} = \frac{1}{3}\text{S}$$

$$u_C(t) = 5 + (4.5 - 5)e^{-3t} = 5 - 0.5e^{-3t} \text{ V}$$

$$i_C(t) = C \frac{du_C}{dt} = 0.5 \times 3 \times 0.5e^{-3t} = 0.75e^{-3t} \text{ A}$$



例6 $t=0$ 时合上开关S。用三要素法求流过开关S的电流*i*。

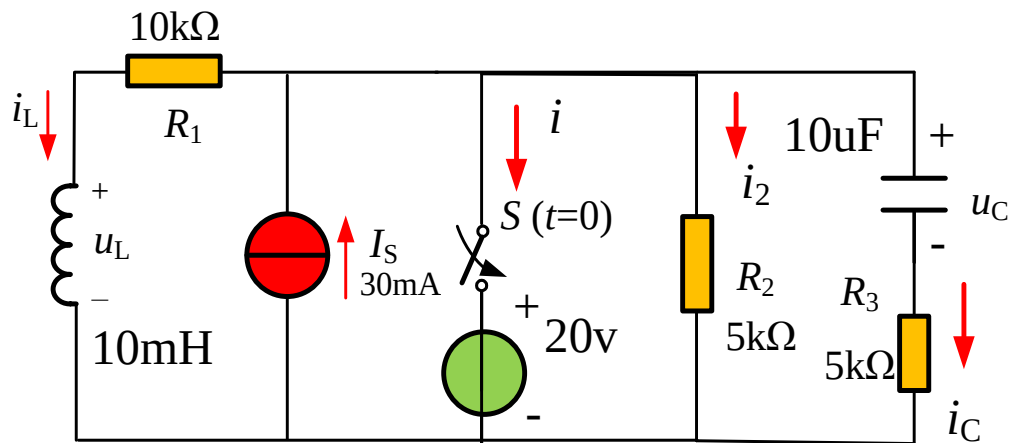
【解】 开关闭S合后电路被分成两个独立部分

(1) 计算电感电流 $i_L(t)$

$$i_L(0_+) = \frac{5}{10+5} \times 30 \times 10^{-3} = 10 \text{mA}$$

$$i_L(\infty) = \frac{20}{10} = 2 \text{mA} \quad \tau = \frac{L}{R} = \frac{10 \times 10^{-3}}{10 \times 10^3} = 10^{-6} \text{S}$$

$$\begin{aligned} i_L(t) &= 2 \times 10^{-3} + (10 \times 10^{-3} - 2 \times 10^{-3}) e^{\frac{-t}{10^{-6}}} \\ &= 2 \times 10^{-3} + 8 \times 10^{-3} e^{\frac{-t}{10^{-6}}} \text{ A } (t \geq 0) \end{aligned}$$



(2) 计算电容电压 $u_C(t)$

$$u_C(0_+) = 5 \times 10^3 \times 20 \times 10^{-3} = 100V \quad u_C(\infty) = 20V$$

$$\tau = 10 \times 10^{-6} \times 5 \times 10^3 = 5 \times 10^{-2} S$$

$$u_C = 20 + (100 - 20)e^{\frac{-t}{5 \times 10^{-2}}} = 20 + 80e^{\frac{-t}{5 \times 10^{-2}}} V$$

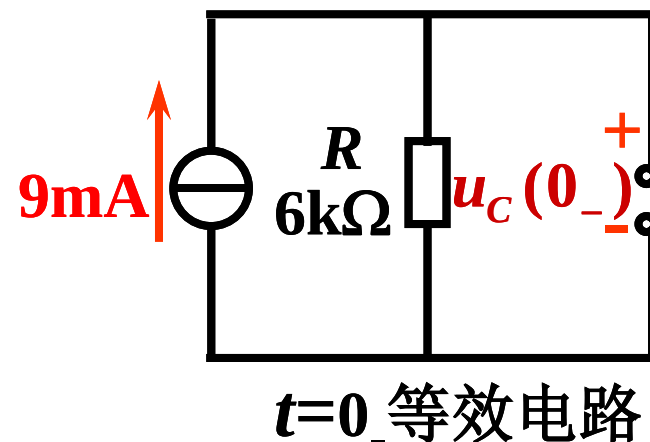
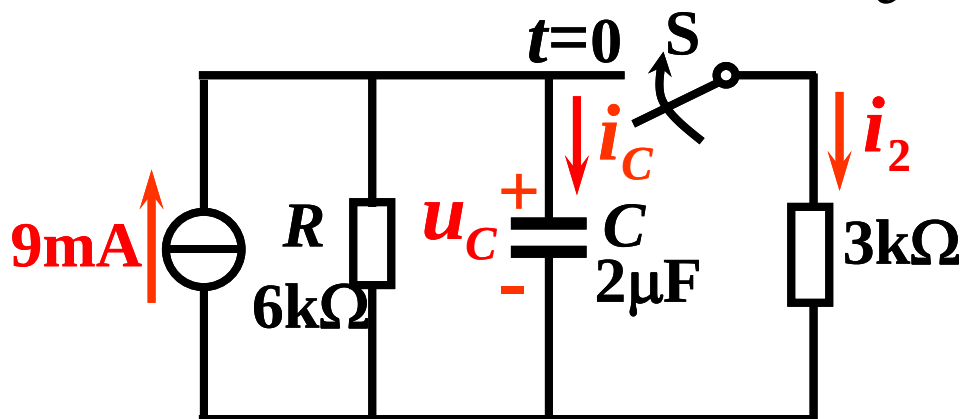
(3) 计算流过开关 S 的电流 i

$$\begin{aligned} i &= 30 \times 10^{-3} - i_L - C \frac{du_C}{dt} - \frac{20}{5 \times 10^3} \\ &= 24 \times 10^{-3} - 8 \times 10^{-3} e^{\frac{-t}{10^{-6}}} - 10 \times 10^{-6} \times \frac{-1}{5 \times 10^{-2}} 80 e^{\frac{-t}{5 \times 10^{-2}}} \\ &= 24 \times 10^{-3} - 8 \times 10^{-3} e^{\frac{-t}{10^{-6}}} + 16 \times 10^{-3} e^{\frac{-t}{5 \times 10^{-2}}} A \end{aligned}$$



例7

电路如图， $t=0$ 时合上开关S，合S前电路已处于稳态。试求电容电压 u_C 和电流 i_2 、 i_C 。



解：用三要素法求解

$$u_C = u_C(\infty) + [u_C(0_+) - u_C(\infty)] e^{-\frac{t}{\tau}}$$

(1) 确定初始值 $u_C(0_+)$

由 $t=0_-$ 电路可求得 $u_C(0_-) = 9 \times 10^{-3} \times 6 \times 10^3 = 54 \text{ V}$

由换路定则 $u_C(0_+) = u_C(0_-) = 54 \text{ V}$



(2) 确定稳态值 $u_c(\infty)$

由换路后电路求稳态值 $u_c(\infty)$

$$u_c(\infty) = 9 \times 10^{-3} \times \frac{6 \times 3}{6 + 3} \times 10^3$$

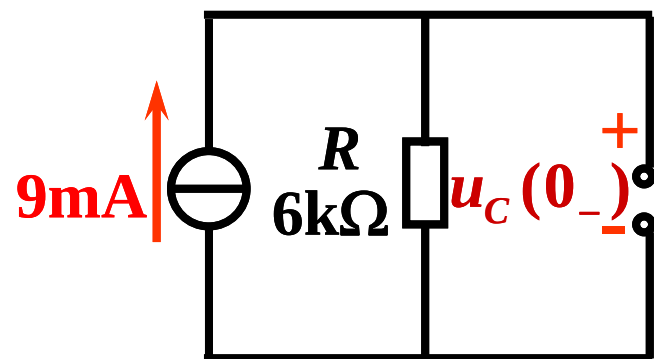
$$= 18 \text{ V}$$

(3) 由换路后电路求 时间常数 τ

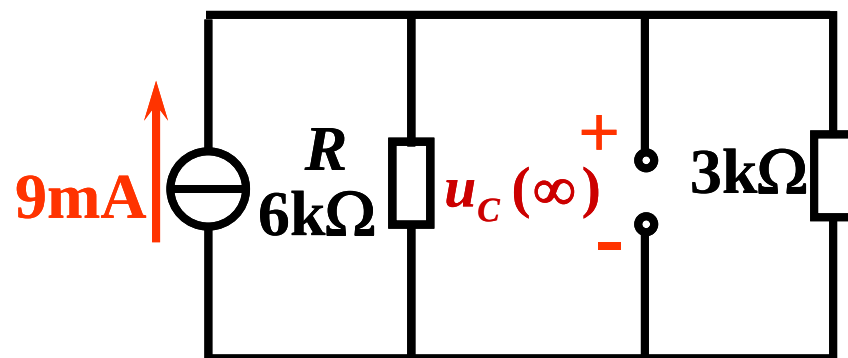
$$\tau = R_0 C$$

$$= \frac{6 \times 3}{6 + 3} \times 10^3 \times 2 \times 10^{-6}$$

$$= 4 \times 10^{-3} \text{ s}$$



$t=0_-$ 等效电路



$t \rightarrow \infty$ 电路

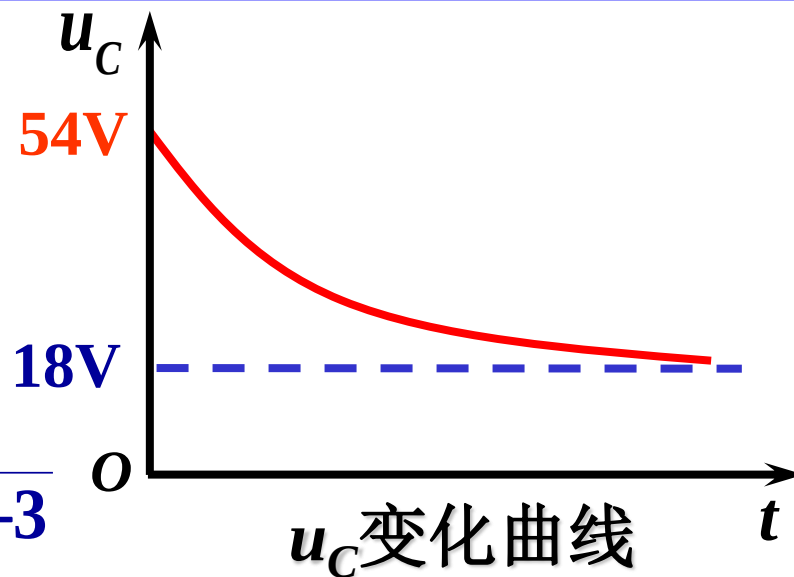


三要素 $\begin{cases} u_C(0_+) = 54 \text{ V} \\ u_C(\infty) = 18 \text{ V} \\ \tau = 4 \times 10^{-3} \text{ s} \end{cases}$

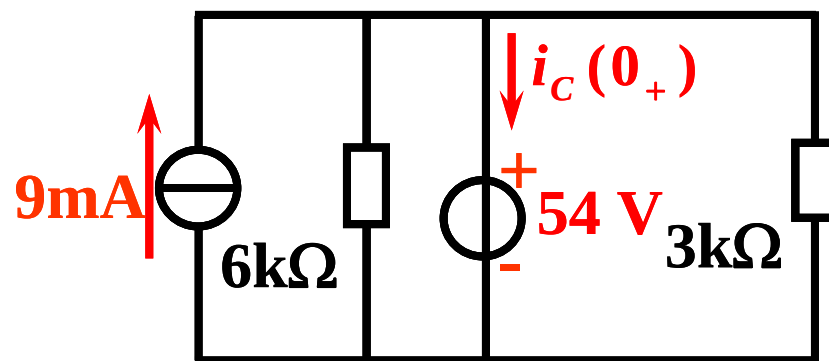
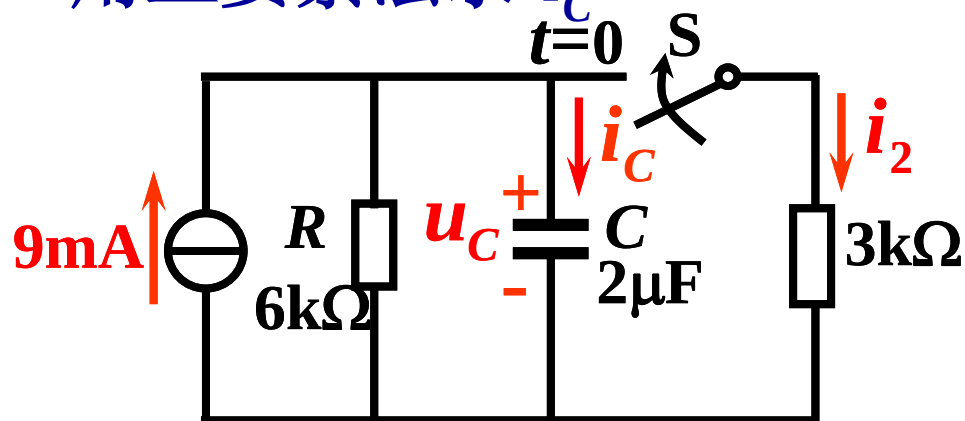
$$\begin{aligned} \therefore u_C &= 18 + (54 - 18)e^{-\frac{t}{4 \times 10^{-3}}} \\ &= 18 + 36e^{-250t} \text{ V} \end{aligned}$$

u_C 的变化曲线如图

$$\begin{aligned} i_C &= C \frac{du_C}{dt} = 2 \times 10^{-6} \times 36 \times (-250)e^{-250t} \\ &= -0.018e^{-250t} \text{ A} \end{aligned}$$



用三要素法求 i_C



$t=0_+$ 等效电路

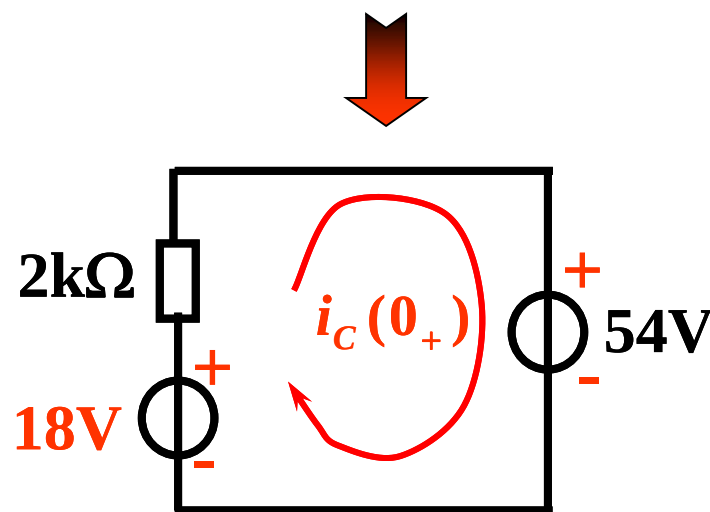
$$i_C = i_C(\infty) + [i_C(0_+) - i_C(\infty)] e^{-\frac{t}{\tau}}$$

$$i_C(0_+) = \frac{18 - 54}{2 \times 10^3} = -18 \text{ mA}$$

$$i_C(\infty) = 0$$

$$i_C(t) = -18e^{-250t} \text{ mA}$$

$$i_2(t) = \frac{u_C(t)}{3 \times 10^3} = 6 + 12e^{-250t} \text{ mA}$$



$t=0_+$



例8

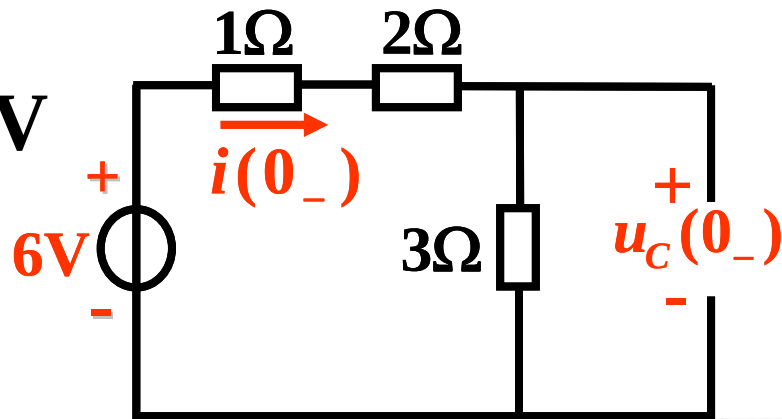
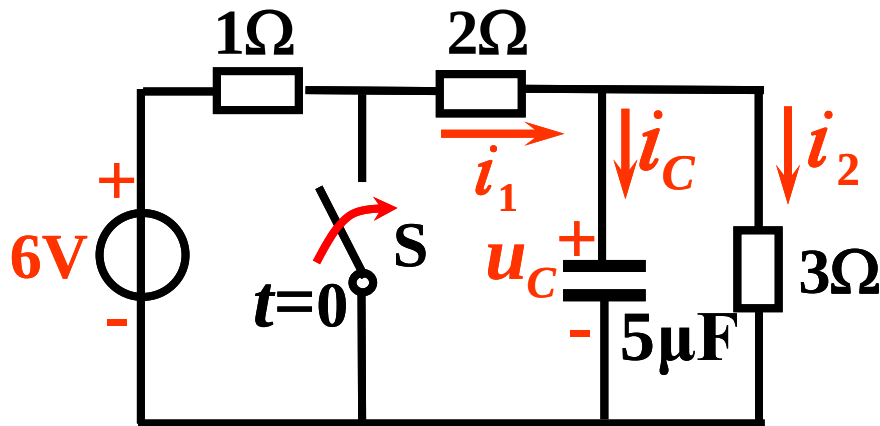
开关S闭合前电路已处于稳态。 $t=0$ 时S闭合，试求： $t \geq 0$ 时电容电压 u_C 和电流 i_C 、 i_1 和 i_2 。

解：用三要素法求解
求初始值 $u_C(0_+)$

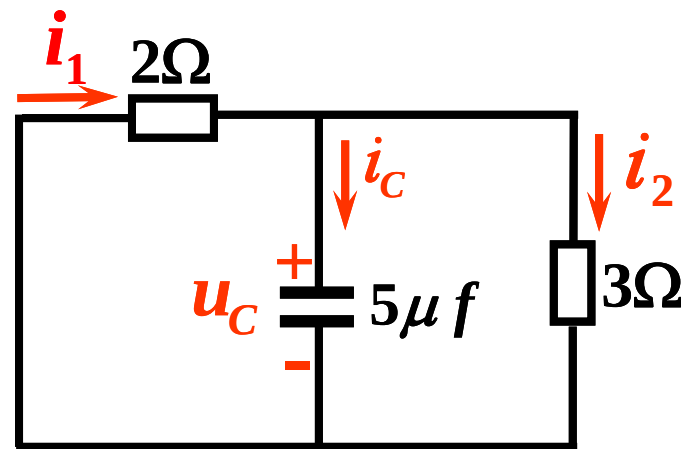
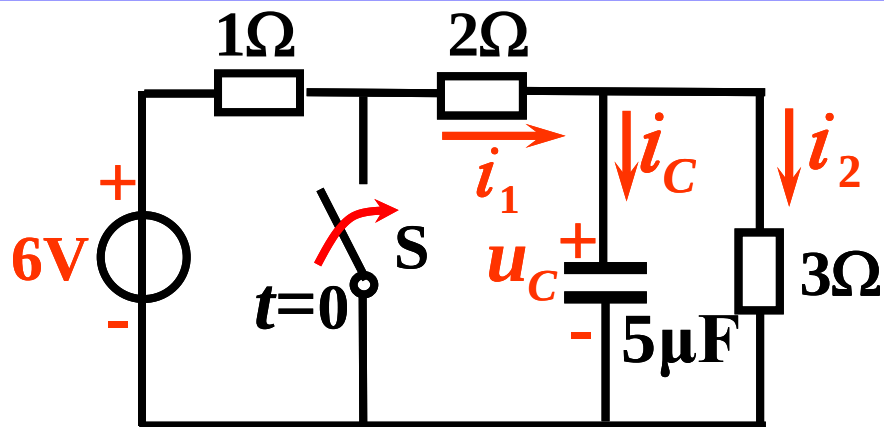
由 $t=0_-$ 时电路

$$u_C(0_-) = \frac{6}{1+2+3} \times 3 = 3 \text{ V}$$

$$u_C(0_+) = u_C(0_-) = 3 \text{ V}$$



$t=0_-$ 等效电路



求稳态值 $u_C(\infty)$ $u_C(\infty) = 0$

求时间常数 τ 由右图电路可求得

$$\tau = R_0 C = \frac{2 \times 3}{2 + 3} \times 5 \times 10^{-6} = 6 \times 10^{-6} \text{ s}$$

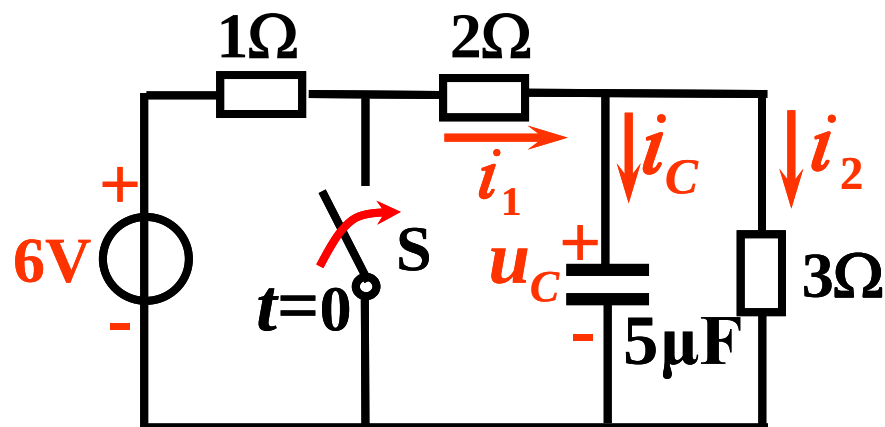
$$\begin{aligned} \therefore u_C(t) &= u_C(\infty) + [u_C(0_+) - u_C(\infty)] U e^{-\frac{t}{\tau}} \\ &= 0 + 3e^{-\frac{10^6}{6}t} = 3e^{-1.7 \times 10^5 t} \text{ V} \end{aligned}$$



$$i_C(t) = C \frac{du_C}{dt}$$

$$= -2.5e^{-1.7 \times 10^5 t} \text{ A}$$

(u_C 、 i_C 关联)



$$i_2(t) = \frac{u_C}{3} = e^{-1.7 \times 10^5 t} \text{ A}$$

$$i_1(t) = i_2 + i_C$$

$$= e^{-1.7 \times 10^5 t} - 2.5e^{-1.7 \times 10^5 t}$$

$$= -1.5e^{-1.7 \times 10^5 t} \text{ A}$$



例9

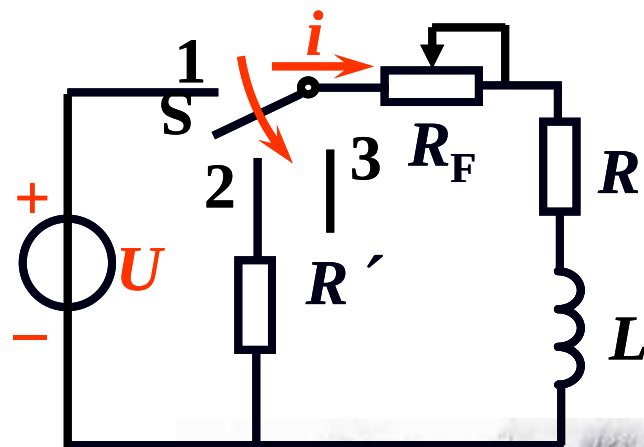
RL 是发电机的励磁绕组，其电感较大。 R_f 用来调节励磁电流。当将电源开关断开时，为了不至由于励磁线圈所储的磁能消失过快而烧坏开关触头，往往用一个泄放电阻 R' 与线圈联接。开关接通 R' 同时将电源断开。经过一段时间后，再将开关扳到3的位置，此时电路完全断开。

已知 $U = 220V$, $L = 10H$, $R = 80\Omega$, $R_f = 30\Omega$ 。

电路稳态时S由1合向2。

(1) $R' = 1000\Omega$, 试求开关S由1合向2瞬间线圈两端的电压 u_{RL} 。

(2) 在(1)中, 若使 U 不超过220V, 则泄放电阻 R' 应选多大?



(3) 根据(2)中所选用的电阻 R' , 试求开关接通 R' 后经过多长时间, 线圈才能将所储的磁能放出95%?

(4) 写出(3) 中 u_{RL} 随时间变化的表示式。

解: 换路前, 线圈中的电流为

$$I = \frac{U}{R + R_F} = \frac{220}{80 + 30} = 2 \text{ A}$$

(1) 开关接通 R' 瞬间线圈两端的电压为

$$u_{RL}(0) = (R_F + R')I = (30 + 1000) \times 2 = 2060 \text{ V}$$

(2) 如果不使 $u_{RL}(0)$ 超过220V, 则

$$(30 + R') \times 2 \leq 220$$

即

$$R' \leq 80 \Omega$$



(3) 求当磁能已放出95%时的电流

$$\frac{1}{2} Li^2 = (1 - 0.95) \frac{1}{2} LI^2$$

$$\frac{1}{2} \times 10 i^2 = 0.05 \frac{1}{2} \times 10 \times 2^2$$

$$i = 0.446 \text{ A}$$

求所经过的时间

$$i = I e^{-\frac{R + R_F + R'}{L} t} = 2e^{-19t}$$

$$0.446 = 2e^{-19t} \quad t = 0.078 \text{ s}$$

$$u_{RL} = -i(R_F + R')$$

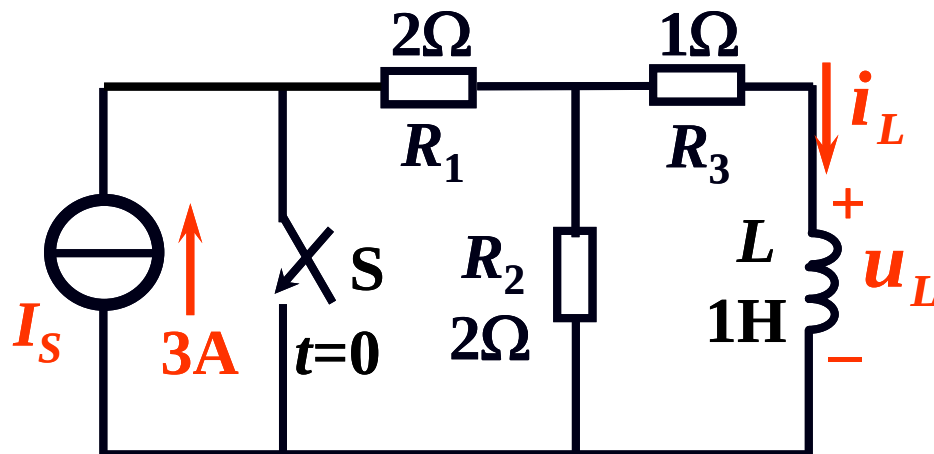
(4) 若按 $R' = 80\Omega$ 计算

$$u_{RL} = -i(30 + 80) = -220e^{-19t} \text{ V}$$



例10

已知S在 $t = 0$ 时闭合，换路前电路处于稳态。
求：电感电流 i_L 和电压 u_L 。



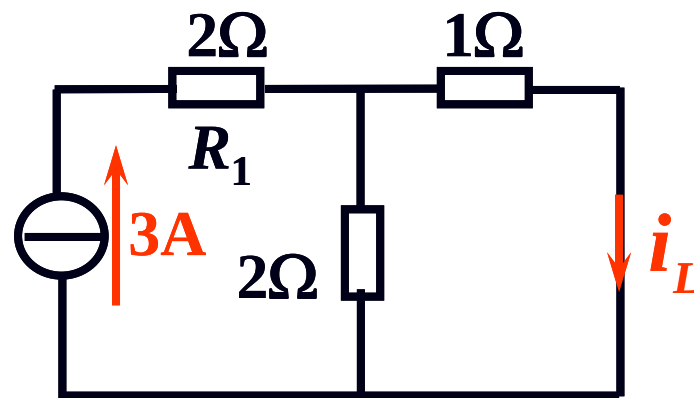
解：用三要素法求解

(1) 求 $u_L(0_+)$, $i_L(0_+)$

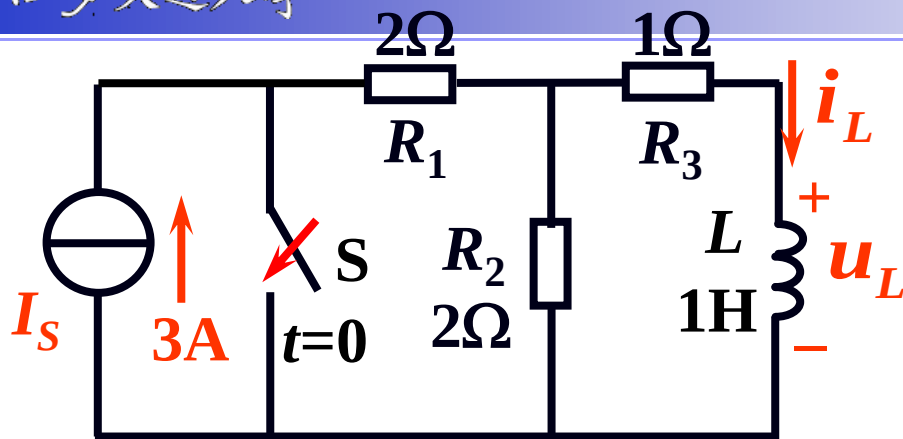
由 $t = 0$ -等效电路可求得

$$i_L(0_-) = \frac{2}{1+2} \times 3 = 2 \text{ A}$$

$$i_L(0_+) = i_L(0_-) = 2 \text{ A}$$



$t = 0$ -等效电路



$$i_L(0_+) = i_L(0_-) = 2 \text{ A}$$

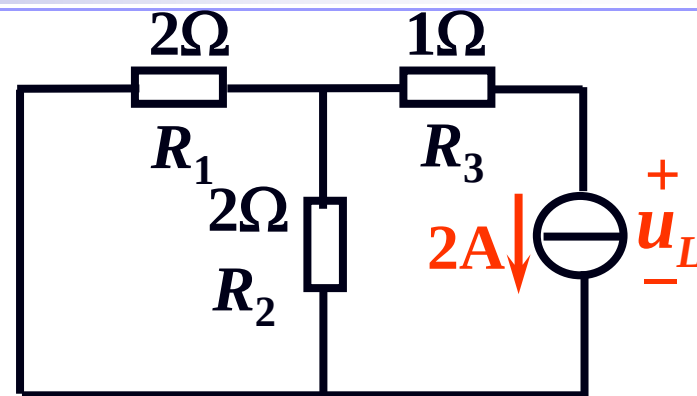
由 $t = 0_+$ 等效电路可求得

$$\begin{aligned} u_L(0_+) &= -i_L(0_+) \times \left(\frac{2 \times 2}{2 + 2} + 1 \right) \\ &= -4 \text{ V} \end{aligned}$$

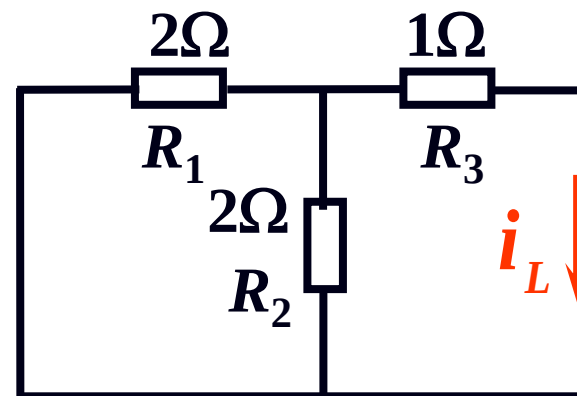
(2) 求稳态值 $i_L(\infty)$ 和 $u_L(\infty)$

由 $t = \infty$ 等效电路可求

$$\text{得 } i_L(\infty) = 0 \text{ V} \quad u_L(\infty) = 0 \text{ V}$$

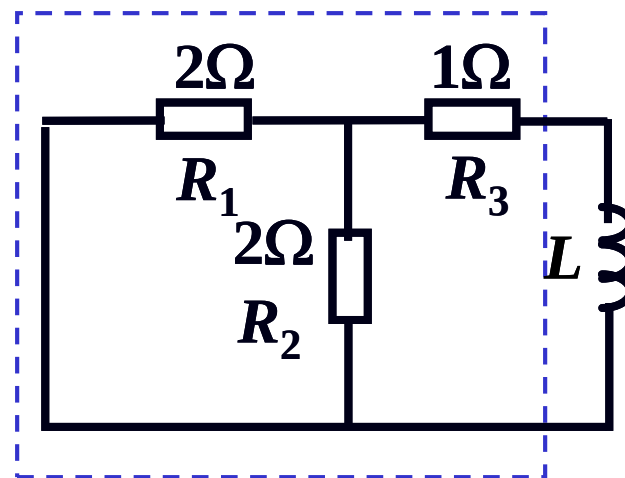
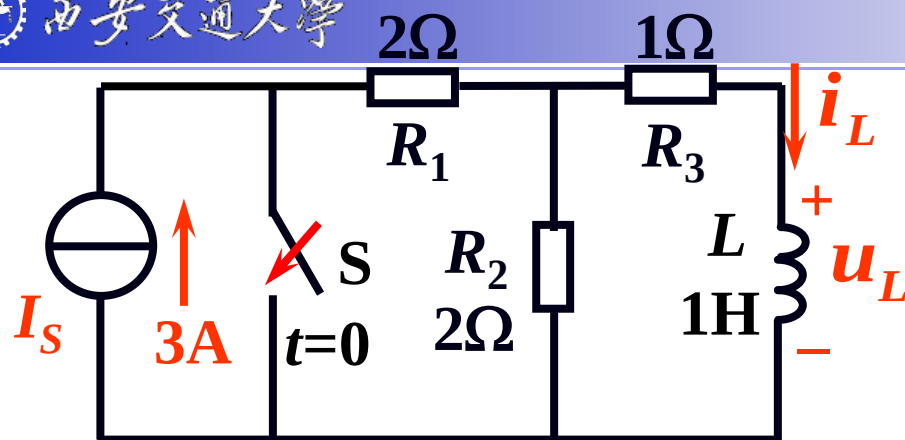


$t = 0_+$ 等效电路



$t = \infty$ 等效电路





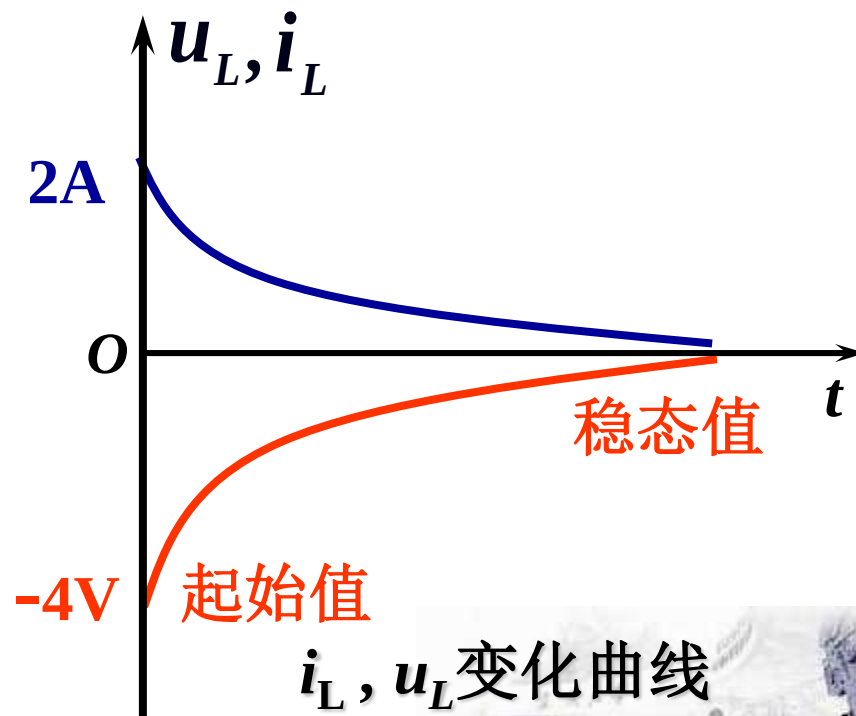
(3) 求时间常数 τ

$$R_0 = R_1 // R_2 + R_3$$

$$\tau = \frac{L}{R_0} = \frac{1}{2} = 0.5 \text{ s}$$

$$i_L = 0 + (2 - 0) e^{-2t} = 2 e^{-2t} \text{ A}$$

$$u_L = 0 + (-4 - 0) e^{-2t} = -4 e^{-2t} \text{ V}$$



例11 电路如图所示，换路前已处于稳态， $t=0$ 时开关S闭合，求 $t \geq 0$ 的 $u_C(t)$ 。

解： 用三要素法求解

1、求初始值 $u_{C(0)}$

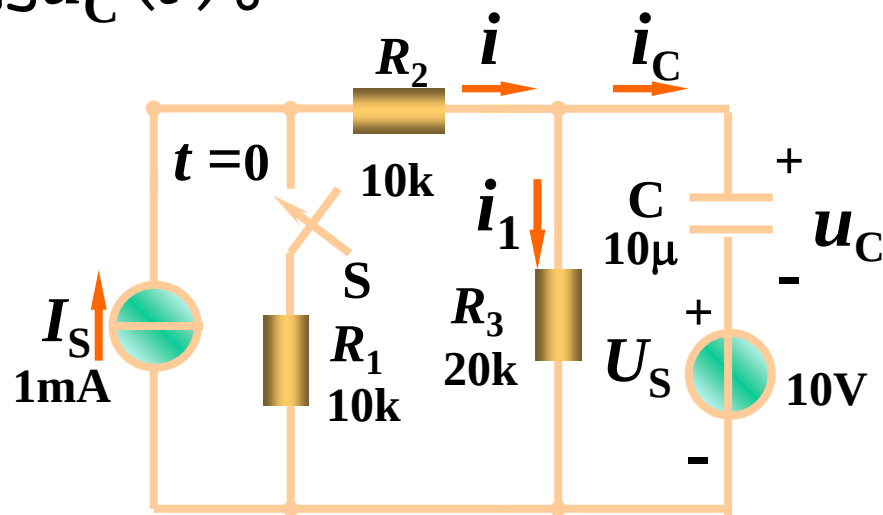
$$u_{C(0)} = R_3 i_1 - U_S$$

$$= 20 \times 10^3 \times 1 \times 10^{-3} - 10$$

$$= 10 \text{ V}$$

2、求稳态值 $u_{C(\infty)}$

$$u_{C(\infty)} = \frac{R_1}{R_1 + R_2 + R_3} I_S R_3 - U_S$$

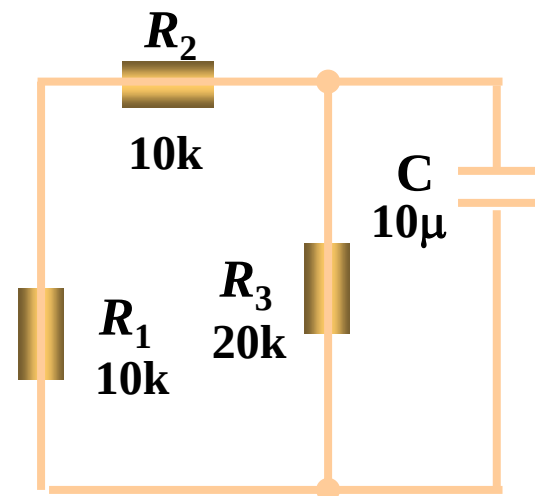
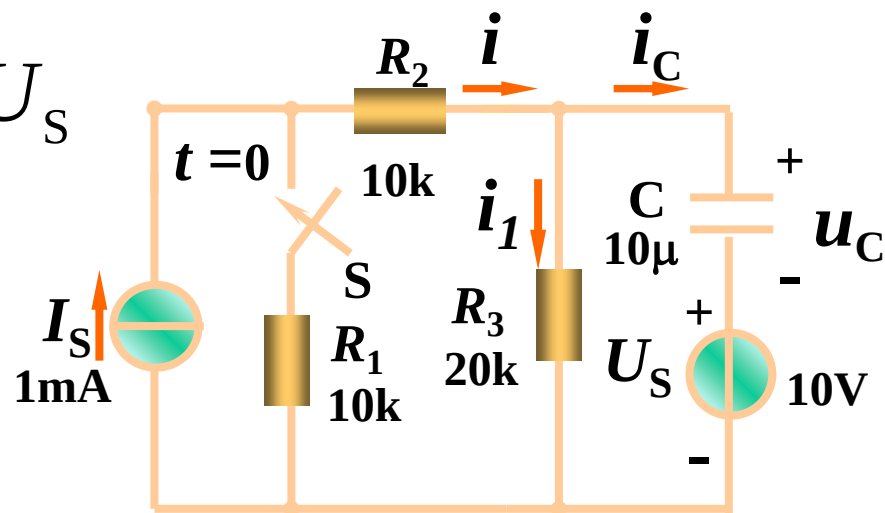


$$\begin{aligned}
 u_{C(\infty)} &= \frac{R_1}{R_1 + R_2 + R_3} I_S R_3 - U_S \\
 &= \frac{1 \times 10 \times 20}{10 + 10 + 20} - 10 \\
 &= -5 \text{ V}
 \end{aligned}$$

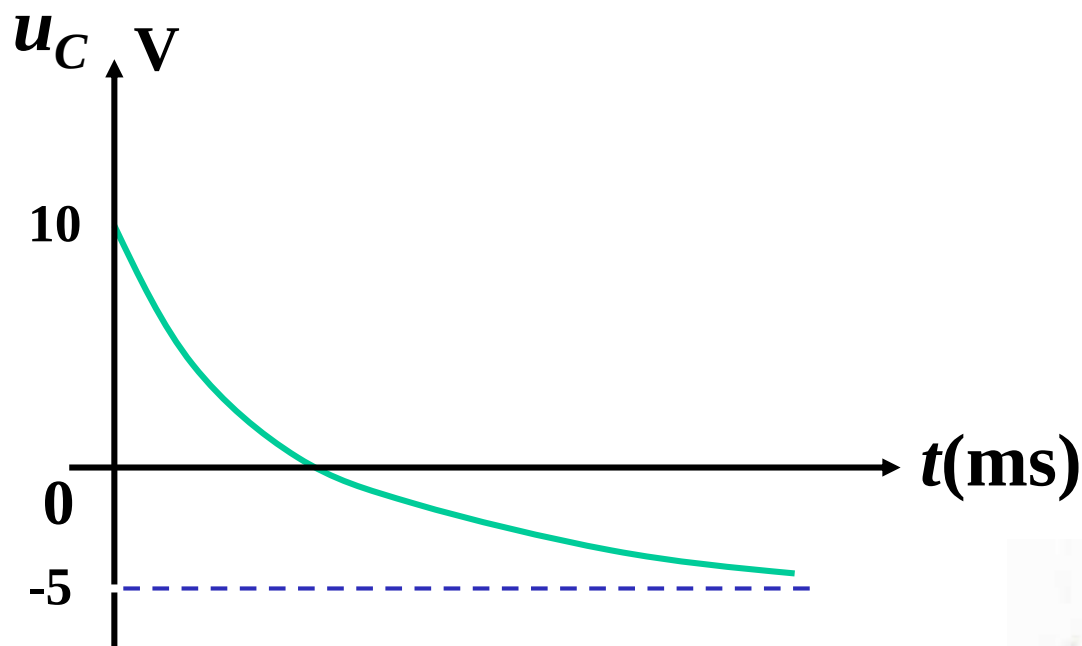
3、求时间常数 τ :

$$\begin{aligned}
 \tau &= [(R_1 + R_2) // R_3] C \\
 &= 100 \text{ ms}
 \end{aligned}$$

$$u_C = u_{C(\infty)} + (u_{C(0)} - u_{C(\infty)}) e^{-t/\tau}$$

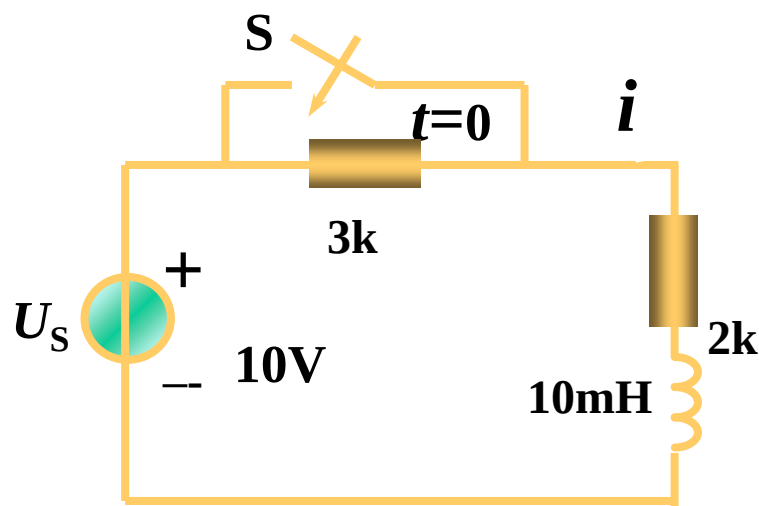


$$\begin{aligned}
 u_C &= u_{C(\infty)} + (u_{C(0)} - u_{C(\infty)})e^{-t/\tau} \\
 &= -5 + (10 + 5)e^{-10t} \\
 &= -5 + 15e^{-10t} \text{ V}
 \end{aligned}$$



例12

S闭合前电路已处于稳态， $t=0$ 时S闭合。试求电流*i*和 $t=5\mu\text{s}$ 时的电流值，画出其波形图？



解： 采用三要素法求 *i*

$$i_{(0)} = \frac{10}{(3+2) \times 10^3}$$

$$= 2 \text{ mA}$$

$$i_{(\infty)} = \frac{10}{2 \times 10^3} = 5 \text{ mA}$$

$$\tau = \frac{L}{R} = \frac{10 \times 10^{-3}}{2 \times 10^3} = 5 \mu\text{s}$$

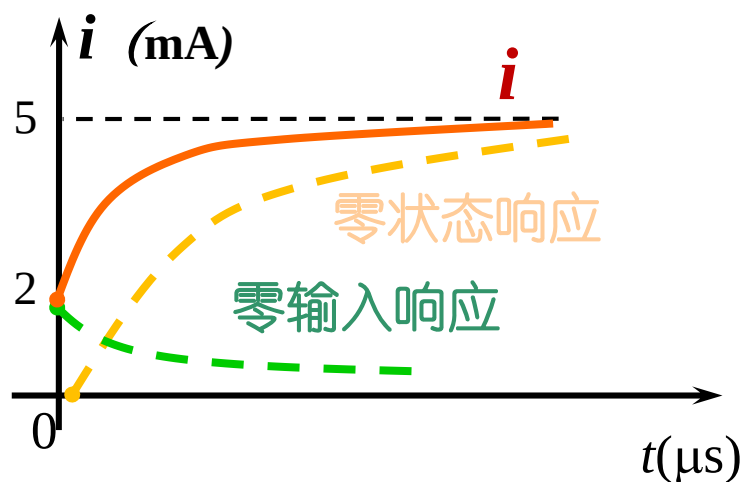
$$i_{(t)} = i_{(\infty)} + (i_{(0)} - i_{(\infty)})e^{-t/\tau}$$



$$i_{(0)} = 2 \text{ mA} \quad i_{(\infty)} = 5 \text{ mA}$$

$$i_{(t)} = i_{(\infty)} + (i_{(0)} - i_{(\infty)})e^{-t/\tau} = 5 - 3e^{-0.2t} \text{ mA}$$

$$i = \text{零输入响应} + \text{零状态响应} = 2e^{-0.2t} + 5(1 - e^{-0.2t})$$

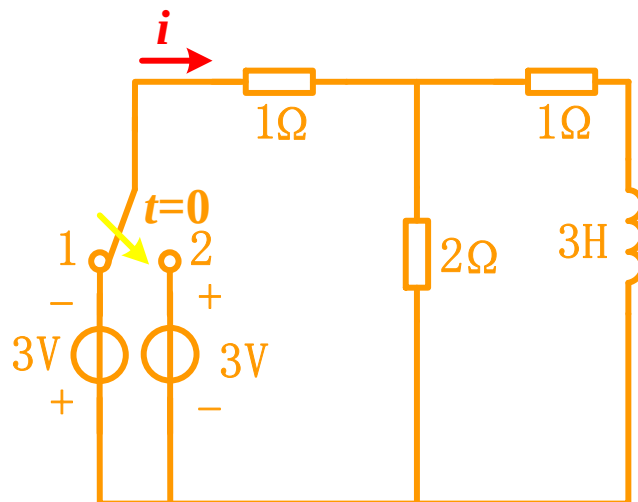


当 $t = 5\mu\text{s}$ 时

$$i(5) = 5 - 3e^{-1} = 3.9 \text{ mA}$$

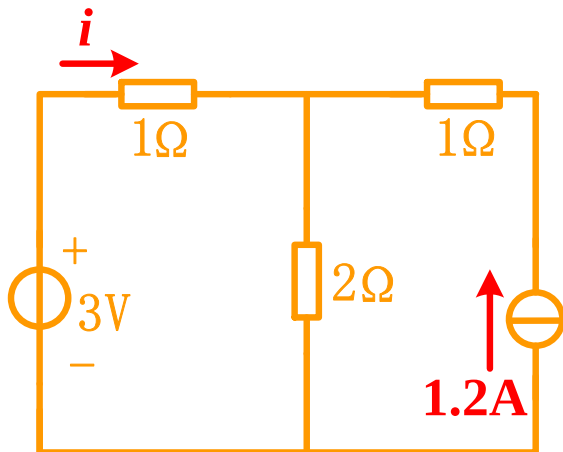


例13 换路前电路已处于稳态, $t = 0$ 时, 开关从1合向2。试求换路后的电流 $i(t)$ 。



解：采用三要素法求 i

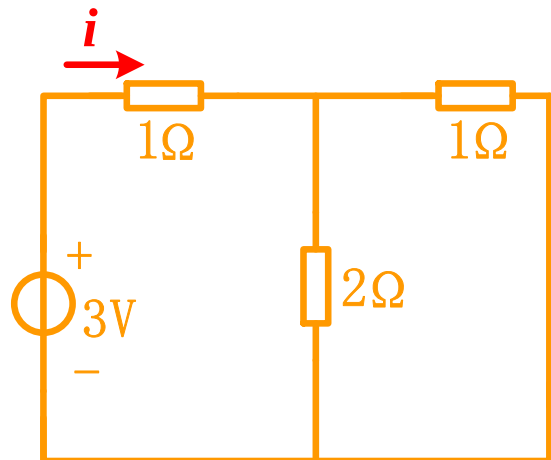
(1) 求 $i(0_+)$, $t=0_+$ 时可以等效为下面的电路。



$$i(0_+) = \frac{3}{3} - \frac{2}{3} \times 1.2 = 0.2 \text{ A}$$

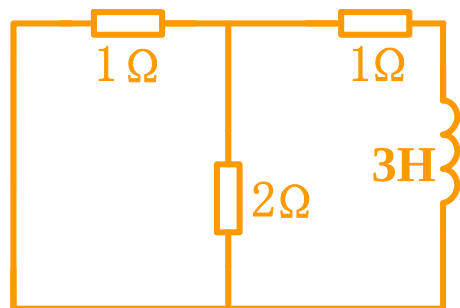


(2) 求 $i(\infty)$, $t=\infty$ 时可以等效为下面的电路。



$$i(\infty) = \frac{3}{1 + \frac{2}{3}} = \frac{9}{5} = 1.8 \text{ A}$$

(3) 求 时间常数 τ

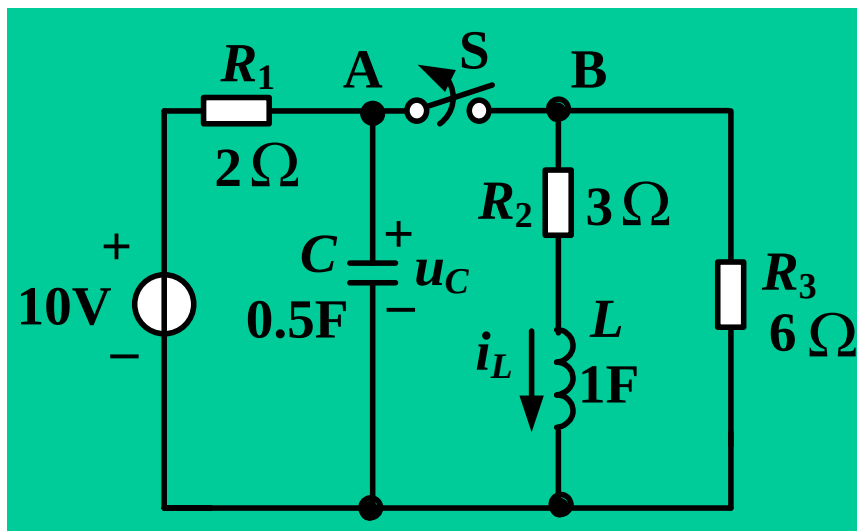


$$\tau = \frac{3}{1 + \frac{2}{3}} = \frac{9}{5} \text{ s}$$

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-\frac{t}{\tau}} = 1.8 + [0.2 - 1.8]e^{-\frac{5t}{9}} = 1.8 - 1.6e^{-\frac{5t}{9}} \text{ A}$$



例14 开关S打开前电路已处于稳态， $t = 0$ 时S打开。
求S打开后的电压 $u_{AB}(t)$ ？



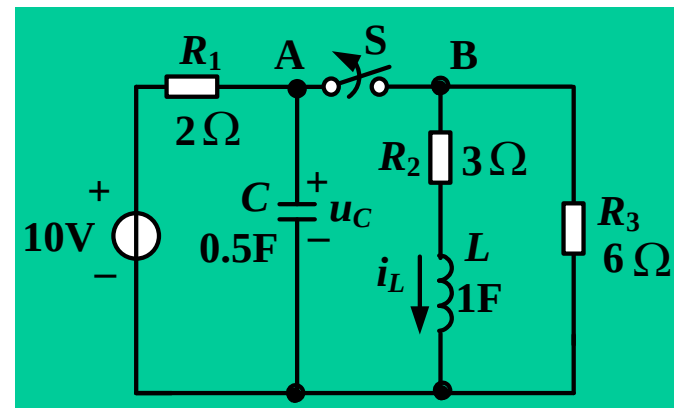
解：开关S打开后电路被分成两个一阶电路，
分别用三要素法求解。



1) 求初始值

$$u_c(0) = \frac{3 // 6}{2 + 3 // 6} \times 10 = 5 \text{ V}$$

$$i_L(0_+) = i_L(0_-) = \frac{u_c(0_-)}{3} = 1.67 \text{ A}$$



2) 求稳态值

$$u_c(\infty) = 10 \text{ V} \quad i_L(\infty) = 0$$

3) 求时间常数, RC 电路部分 $\tau_1 = R_1 C = 1 \text{ s}$

$$RL \text{ 电路部分} \quad \tau_2 = \frac{L}{R_2 + R_3} = \frac{1}{9} \text{ s}$$



4) 写出 $u_c(t)$ 与 $i_L(t)$ 的表达式

$$u_c(t) = u_c(\infty) + [u_c(0_+) - u_c(\infty)]e^{-\frac{t}{\tau_1}}$$

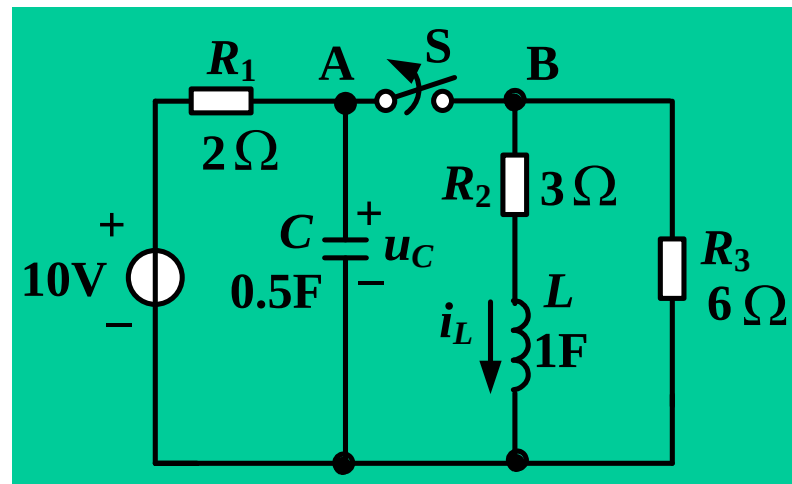
$$= 10 - 5e^{-t} \text{ V}$$

$$i_L(t) = i_L(\infty) + [i_L(0_+) - i_L(\infty)]e^{-\frac{t}{\tau_1}} = 1.67e^{-9t} \text{ A}$$

所以

$$u_B(t) = u_{R3}(t) = -R_3 i_L(t) = -10e^{-9t} \text{ V}$$

$$u_{AB}(t) = u_c(t) - u_{R3}(t) = 10 - 5e^{-t} + 10e^{-9t} \text{ V}$$





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谢谢